Subchat Trees: Architecture Overview

a Interactive Branching Interface

User-driven branching from selected parent text

Selection-Based Branching

- User highlights a text span in the parent node
- A focused child subchat is created from that span
- Follow-up prompt F is attached to guide the child
- Parent context is inherited once at creation

Tree Navigation View

- Root node represents the main conversation
- Child nodes represent focused follow-up threads
- Depth encodes lineage r → ... → n
- Sibling branches remain context-isolated

Messenger-Style Interaction — Familiar chat interface with in-place branching and focused follow-ups

Branching preserves the parent thread while opening a new child context.

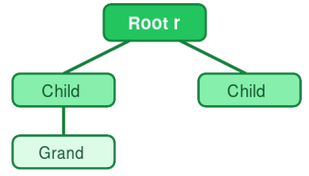
(a) User interface with text-selection-driven subchat creation

creates

b Hierarchical Conversation Forest

Forest = {T1, T2, ..., Tp} of rooted conversation trees

Tree T = (N, E, r, B, V)



Path: r → Child → Grand

Node Semantics

- Each node stores title, parent, children, and path
- Each node owns a local buffer B_n
- Follow-up nodes store selected-text context
- Child nodes copy parent state once at creation
- Subsequent turns remain local to that branch

Def. 1. *Each node n ∈ N has local state, parent linkage, and branch-specific memory. Siblings do not share context; inherited occurs only at child creation.*

(b) Hierarchical tree structure for conversation organization

LLM Response

responds with prior knowledge

R (retrieved) (on-demand)

LLM

B_n + S_sum (always)

isolates

d On-Demand Retrieval Pipeline

Long-term retrieval from the vector archive when local context is insufficient

Retrieval trigger: The system decides whether archived context is needed for the current query
If not triggered, generation proceeds using node-local memory only.

Phase 1 Decompose
Generate q plus up to n ≤ 8 focused reformulations of the user query

Phase 2 Search
Embed each sub-query with all-mpnet-base-v2 and search ChromaDB (N = 20 candidates/query)

Phase 3 Re-rank
Deduplicate the candidate pool
Re-rank against the original query and keep top-k anchors

Phase 4 Expand
Expand each anchor into a turn-based window ±δ (δ = 2) to form coherent context blocks R

Embedding model: *all-mpnet-base-v2, d = 768*

Output: a small set of coherent retrieved blocks R for prompt assembly

(d) Multi-query retrieval pipeline for on-demand long-term memory access

queries

c Three-Tier Node Memory

Per-node short-, medium-, and long-term context

Tier 1 Short-Term: Circular Buffer B_n
FIFO deque · C messages (default 15) · Per-node isolated

Tier 2 Medium-Term: Rolling Summary S_n
S_n = LLM(S_{n−1} ⊕ M_buffer) · Triggered at C, 2C, 3C, ...

Tier 3 Long-Term: Vector Archive V
Immediate vector indexing with node_id, role, timestamp, and turn number

Prompt assembly: local buffer and rolling summary are always available
Retrieved blocks R are appended only when retrieval is triggered for the current query.

Context Inheritance (Eq. 4):
At child creation: selected-text prompt + parent summary + parent buffer copy
One-time inheritance · No sibling sharing · No backward propagation to the parent

(c) Local buffer with fixed-size memory and rolling summarization